Decision Models in GeoAI:

Testing Jev for qualifying spatial relations.

Master's Thesis

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Abstract

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Acronyms

AI artificial intelligence. 2

GIS geoinformation systems. 2

LLMs large language models. 2

1. Introduction

Janowicz et al., 2020 asked if there will be an *"artificial GIS analyst"* by 2030. Since then there have been huge strides both in the field of artificial intelligence (AI) and autonomous geoinformation systems (GIS) (Li et al., 2025). But using large language models (LLMs) for spatial analysis requires large amounts compute, leading to low accessibility and high environmental impact (Janowicz, 2023; Shi et al., 2025). LLMs may produce unnecessary reasoning tokens with high latency (Chen et al., 2025; Duan et al., 2026; Wang et al., 2026) and a long response in natural language, which is difficult to process downstream. *Decision models*, also described as *system-one* models (Kahneman, 2011), might be more suitable for certain GeoAI (Janowicz et al., 2020) applications, as they operate more efficiently and quickly than LLMs, as well as responding in a predetermined, consistent format. To test this assumption, this thesis reproduces the experiment on *spatial relation qualification* done by Ji et al., 2025 using Typesafe AI's model Jev (Almeida, 2026).

2. Research Goals

3. Related Works

3.1 GeoAI

3.2 General versus Decisional Intelligence

This decade has seen the emergence of AI from highly specialized models to task-agnostic LLMs, that show general intelligence or "*the ability to adapt across diverse tasks and domains without task-specific retraining*" (Duan et al., 2026, p.2). As frontier models grew in size and became more sophisticated, their abilities in text and image generation, software development, information retrieval, and reasoning improved. Alongside these advances, the need for compute and price of accessing frontier models is rising steadily.

With the recent release of the model Jev by the company Typesafe AI (Almeida, 2026) a new type of AI model, dubbed *general decision models* (Duan et al., 2026), was introduced. Specialized on *decisional intelligence*, or quick "off-the-cuff" thinking akin to *system-one intelligence* (Kahneman, 2011), decision models prioritize quick, efficient responses over in-depth reasoning. Rather than taking a prompt in natural language and generating an open-ended response, decision models receive structured instructions to choose the answer between predetermined options.

3.3 Benchmarking Jev

3.4 Ethical Remarks to GeoAI

3.5 Qualifying Spatial Relations

4. Experiment

4.1 Data

4.2 Prompts

4.3 Results

5. Discussion

6. Conclusion

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Appendix

Use of Artificial Intelligence

For this thesis, Anthropic Claude Opus 5.5 [66] was employed to assist in refining and improving the text throughout all sections of this paper. This model was also used to help write the code for this paper, which is available at github.com/AppliedMachineLearn Lab/jev-benchmarking. The authors retain full responsibility for the accuracy, integrity, and originality of the work. (Template from Deußer et al., 2026)